

Rushil Krishna Sai Narendula

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EDUCATION

Indian Institute of Technology Dharwad

Jul 2024 – May 2028

B.Tech in Mathematics and Computing; CGPA: 9.64/10

- **Relevant Coursework:** Probability(10) & Statistics(10), Linear Algebra(AP), DBMS, Data Structures & Algorithms, Design Analysis And Algorithms(AP), Numerical Methods, Data Analysis(10)

TECHNICAL SKILLS

Languages: Python, SQL, C++, C, JavaScript

Data & Analysis: Pandas, NumPy, Statistical Analysis, Data Visualization, ETL Pipelines, EXPLAIN/ANALYZE Query Profiling, Bayesian Optimization using Random Forest and Gaussian Method

ML: Supervised Learning, Unsupervised Learning, Neural Networks (CNNs, RNNs), Scikit-learn, pytorch

Databases: PostgreSQL, MongoDB

Tools: Git, GitHub, Linux, VS Code, Postman

DATA & RESEARCH EXPERIENCE

Research Intern – Data Systems & Query Workload Analysis

May 2026 – Present

Indian Institute of Technology Hyderabad | [Code](#)

- Replicated and critically evaluated **SQLBarber**, an LLM-driven SQL workload synthesis pipeline, generating and analyzing 113 synthetic queries against real workloads on the Join Order Benchmark (JOB/IMDB) dataset
- Profiled cost distributions across 33 base query templates and found **81.8% (27/33)** drifted into unintended bins, exposing a statistical fidelity gap between synthetic and real-world query data
- Designed a custom **Query Relative Ratio (QRR)** metric and built 10+ interactive dashboards to quantify and visualize per-query prediction error across all 113 query pairs
- Engineered a Python (psycopg2) data pipeline to extract, parse, and diff PostgreSQL EXPLAIN/ANALYZE execution plans at scale, structuring raw query-plan output into analyzable datasets
- Applied Bayesian Optimization (Random Forest and Gaussian Process surrogates) to iteratively refine workload generation parameters across 19 query templates, improving cost-distribution alignment post-refinement

Research Intern – Mathematical Foundations of Machine Learning

Dec 2024 – Jan 2025

Indian Institute of Technology Indore

- Implemented matrix decomposition techniques (Gram-Schmidt, QR) and validated core ML algorithms (KNN, SVM, linear/logistic regression) against expected outputs; certificate of completion awarded

PROJECTS

Facial Recognition under Varying Illumination | *PyTorch, Scikit-learn, OpenCV, PCA*

[Code](#)

- Benchmarked 5 modeling approaches on a 16,000+ image dataset spanning 28 subjects; fine-tuned **ResNet50** (transfer learning, label smoothing, cosine annealing) reached **98.13% accuracy**, outperforming a custom 5-block CNN (96.13%)
- Built PCA-based Eigenface dimensionality reduction (450 components, 95% variance retained) for classical classifiers, achieving 91.06% accuracy with SVM (RBF kernel) and 83.15% with KNN (k=1)
- Tuned a Random Forest classifier via RandomizedSearchCV on raw high-dimensional pixel data (73.34% accuracy), analyzing why PCA reduction degraded tree-based performance versus distance- and margin-based classifiers
- Built a FaceNet embedding pipeline with CLAHE and gamma-correction preprocessing to recover facial features under heavy shadow occlusion, fine-tuning the top 20 layers of a pretrained backbone

UNI HUB – University Community Platform | *Node.js, Express.js, MongoDB, JWT, OAuth*

[Demo](#)

- Implemented JWT and Google OAuth authentication with role-based access control to secure multi-user access across the platform
- Designed MongoDB schemas and REST APIs to support threaded discussions, nested comments, and permission management for a production-deployed platform

TravelNext – Tourist Listing and Review Platform | *Node.js, Express, MongoDB, EJS, Joi*

[Code](#) | [Demo](#)

- Built a full-stack MVC application with RESTful CRUD architecture and independently deployed it to Render with MongoDB Atlas
- Designed centralized Joi validation middleware and error-handling logic to enforce data integrity across API endpoints

COMPETITIVE PROGRAMMING

- Solved 650+ algorithmic problems (470+ LeetCode, rating 1650+; 180+ Codeforces, rating 1056), with contest-tested experience in graph algorithms, dynamic programming, and greedy algorithms

LEADERSHIP & ACHIEVEMENTS

- **Head, Competitive Programming Wing, Coding Club, IIT Dharwad** – Organize daily DSA practice, contest mashups, and mentor 100+ juniors
- 3rd Place, DevHack 7.0 Hackathon (40+ teams); Participant, Smart India Hackathon (SIH) 2025
- 6 AP (Highest Distinction) grades, including Design & Analysis of Algorithms and Linear Algebra; Member, Quiz Club